

## Hawkwood Pond & Wetlands Ecological Sewage System 2010



**Hawkwood College** is an independent adult education centre offering courses in creative exploration to people of all ages and from all walks of life. The College seeks to foster a spiritual foundation of life. Its ethos includes a commitment to sustainable land use, hence the decision to build a state-of-the-art ecological sewage system which would enhance the grounds and enrich the natural ecology.

**How does the sewage system work?** Sewage and waste water from the house (bathrooms, toilets, kitchen and laundry but not rain water) is piped into two digestion tanks (right), where solids settle and separate. The liquids flow into a distribution chamber which then supplies each of the four primary treatment ponds with 60 litres around ten times a day. The College produces the equivalent organic load to 40 permanent residents.



**A river ecology.** In each pond a pump recirculates the water through a cascade and marginal beds of aquatic plants. Three of the cascades are Flowform designs with rhythmical vortex movement and the fourth is a naturalistic stream. Each pond recreates a river ecology in a different way and makes use of the water cleaning process found in a watercourse. Cleansed overflow from the primary treatment ponds joins together in a fifth maturation pond,

also with planted gravel beds, but no cascade. After this comes the water absorption area, in the form of a meandering ditch planted with willows, dogwood and other water-hungry plants. This area allows a final polishing of the water before released into the environment.

**Biodiversity.** The whole system works as a complete organism set into the ecology of the site. It has had a wonderful impact on the local biodiversity supporting wildfowl, grass snakes, amphibians and a wide range of smaller aquatic animals. The ponds have also proved popular with residents and visitors who walk around and rest by them.



**Design.** The system is regularly observed and monitored to gain further insight into the function of a natural water cleaning system and the way Flowforms support the ecology. It was designed and built by Simon Charter and Andy Jones of Ebb & Flow Ltd, Nailsworth (Simon seen here on the right with Hawkwood Patron Jonathon Porritt). The project won a 2005 Stroud District Council Award for Sustainable Design.

More information from: Ebb & Flow Ltd, Ruskin Mill, Old Bristol Road, Nailsworth, Gloucestershire, GL6 0LA  
Tel: 01453 836060 Email: [info@ebbandflowltd.co.uk](mailto:info@ebbandflowltd.co.uk) Website: [www.ebbandflowltd.co.uk](http://www.ebbandflowltd.co.uk)